CASE 2
CD40 LIGAND DEFICIENCY

Failure of immunoglobulin class switching.

After exposure to an antigen, the first antibodies to appear are IgM. Later, antibodies of other classes appear: IgG predominates in the serum and extravascular space, while IgA is produced in the gut and in the respiratory tract, and IgE may also be produced in the mucosal tissues. The different effector functions of these different antibody classes are summarized in Fig. 2.1. The changes in the class of the antibody produced in the course of an immune response reflect the occurrence

Functional activity	IgM	IgD	IgG1	IgG2	IgG3	IgG4	IgA	IgE
Neutralization	+	-	++	++	++	++	++	-
Opsonization	-	-	+++	*	++	+	+	-
Sensitization for killing by NK cells	-	-	++	-	++	-	-	-
Sensitization of mast cells	-	-	+	-	+	-	-	+++
Activation of the complement system	+++	-	++	+	+++	-	+	-
Distribution	**IgM**	**IgD**	**IgG1**	**IgG2**	**IgG3**	**IgG4**	**IgA**	**IgE**
Transport across epithelium	+	-	-	-	-	-	+++ (dimer)	-
Transport across placenta	-	-	+++	+	++	+/-	-	-
Diffusion into extravascular sites	+/-	-	+++	+++	+++	+++	++ (monomer)	+
Mean serum level (mg ml^{-1})	1.5	0.04	9	3	1	0.5	2.1	3×10^{-5}

Fig. 2.1 **Each human immunoglobulin isotype has specialized functions and a unique distribution.** The major effector functions of each isotype (+++) are shaded in dark red, while lesser functions (++) are shown in dark pink, and very minor functions (+) in pale pink. The distributions are similarly marked, with the actual average levels in serum of adults shown in the bottom row. *IgG2 can act as an opsonin in the presence of Fc receptors of a particular allotype, found in about 50% of Caucasians.

TOPICS BEARING ON THIS CASE:

- Isotype or class switching
- Antibody isotypes and classes
- CD40 ligand and class switching
- Antibody-mediated bacterial killing

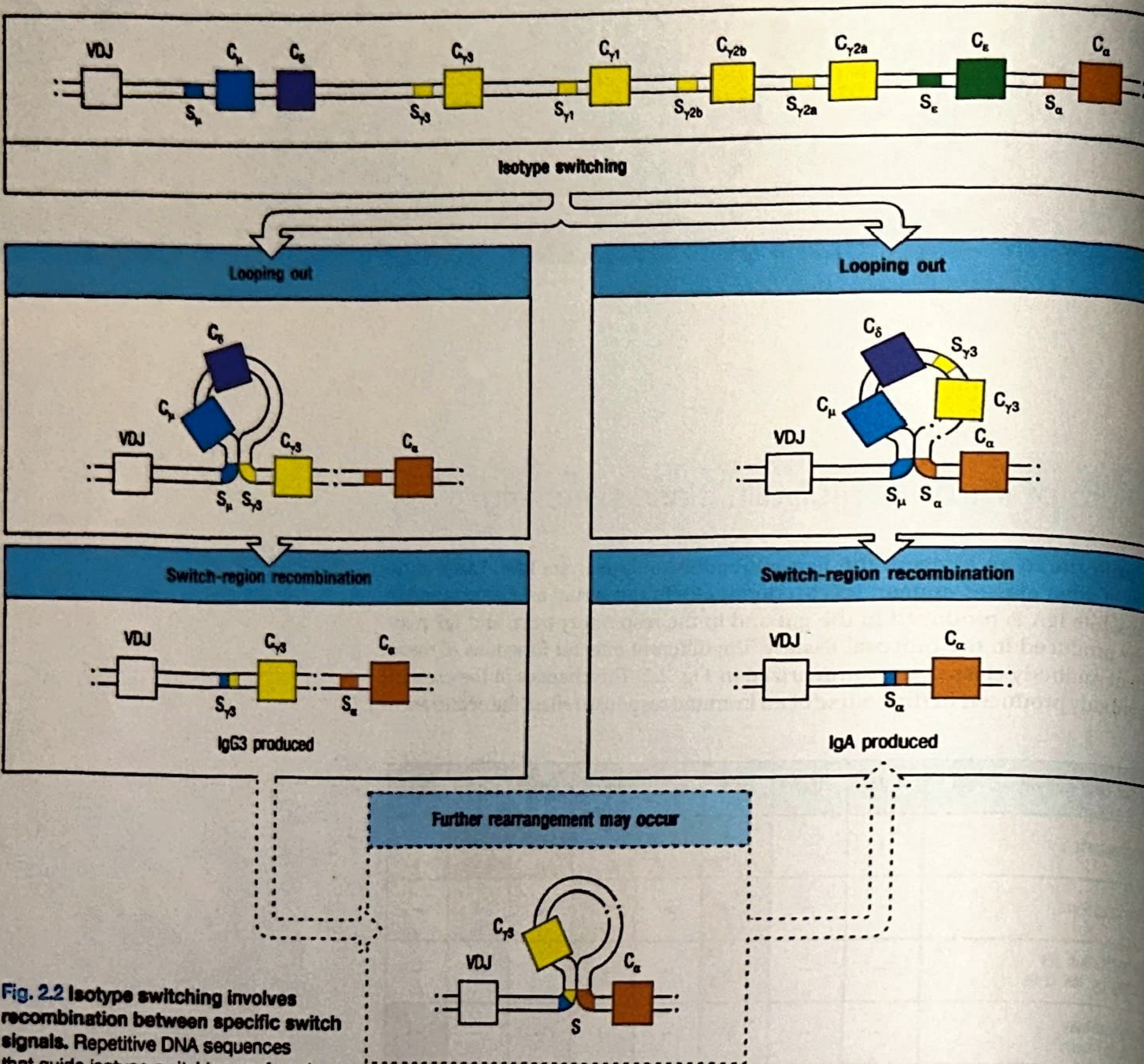

Fig. 2.2 Isotype switching involves recombination between specific switch signals. Repetitive DNA sequences that guide isotype switching are found upstream of each of the immunoglobulin C-region genes (in this case, the mouse heavy-chain locus), with the exception of the C_δ gene. Switching occurs by recombination between these repetitive sequences or switch signals as a result of the repair of double-strand breaks (see Case 3), with deletion of the intervening DNA. The initial switching event takes place from the μ switch region (Sμ): switching to other isotypes can take place subsequently from the recombinant switch region formed after μ switching. S, switch region.

of heavy-chain isotype switching in the B cells that synthesize immunoglobuli so that the heavy-chain variable (V) region, which determines the specificity of a antibody, becomes associated with heavy-chain constant (C) regions of differen isotypes, which determine the class of the antibody, as the immune response pro gresses (Fig. 2.2).

Class switching in B cells, also known as isotype switching and class-switch recom bination, is induced mainly by T cells, although it can also be induced by T-cel independent Toll-like receptor (TLR)-mediated signaling. T cells are required t initiate B-cell responses to many antigens; the only exceptions are responses trig gered by some microbial antigens or by certain antigens with repeating epitopes This T-cell 'help' is delivered in the context of an antigen-specific interaction with the B cell (Fig. 2.3). The interaction activates the T cell to express the cell-surface protein CD40 ligand (CD40L, also known as CD154), which in turn delivers an activating signal to the B cell by binding CD40 on the B-cell surface. Activated T cells secrete cytokines, which are required at the initiation of the humoral immune response to drive the proliferation and differentiation of naive B cells, and are later required to induce class switching (Fig. 2.4). In particular, follicular helper T cells (T_{FH}) produce interleukin-21 (IL-21), which promotes the germinal cente

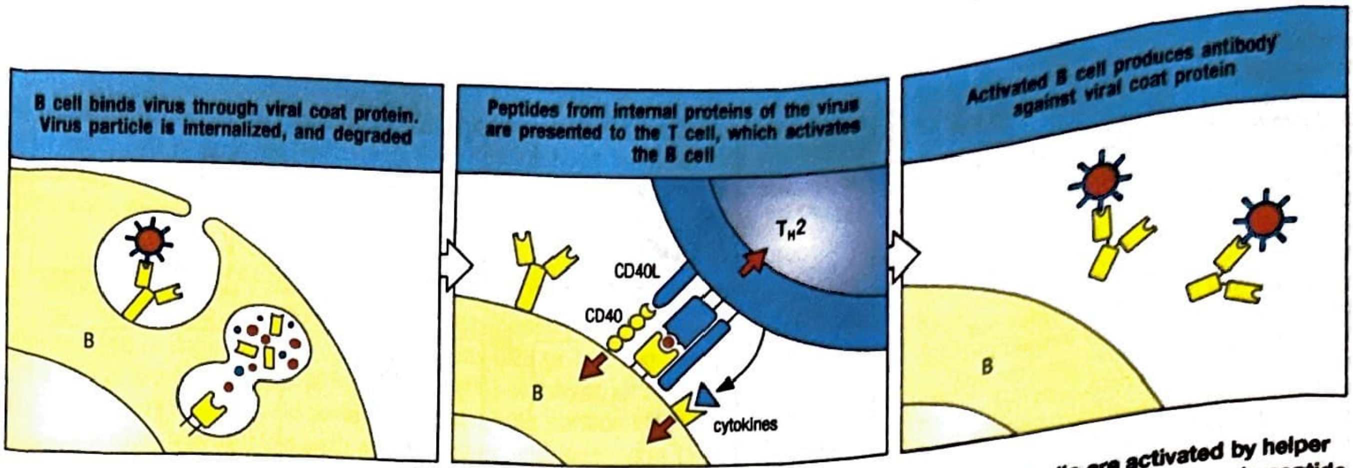

Fig. 2.3 B cells are activated by helper T cells that recognize antigenic peptide bound to class II molecules on their surface. An epitope on a viral coat (spike) protein is recognized by the surface immunoglobulin on a B cell, and the virus is internalized and degraded. Peptides derived from viral proteins are returned to the B-cell surface bound to MHC class II molecules, where they are recognized by previously activated helper T cells that activate the B cells to produce antibody against the virus.

reaction and differentiation of B lymphocytes into plasmablasts. Other cytokines induce class switch recombination of B cells. In humans, class switching to IgE synthesis is best understood, and is known to require interleukin-4 (IL-4) or IL-13, as well as stimulation of the B cell through CD40.

The gene for CD40L (*CD40LG*) is located on the X chromosome at position Xq26. In males with a defect in this gene, isotype switching fails to occur; such individuals make only IgM and IgD and are severely impaired in their ability to switch to IgG, IgA, or IgE synthesis. This phenotype is known generally as 'hyper IgM syndrome,' and can also be due to defects other than the absence of CD40L (see Case 3). Similarly, defective class switching is also observed in patients with CD40 deficiency, a rare autosomal recessive condition. Defects in class switching can be mimicked in mice in which the genes for CD40 or CD40L have been disrupted by gene targeting; B cells in these animals fail to undergo switching. The underlying defect in patients with CD40L deficiency can be readily demonstrated by isolating their T cells and challenging them with soluble, fluorescently labeled CD40 (made by engineering the extracellular domain of CD40 onto the constant region (Fc) of IgG) or with monoclonal antibodies that recognize the CD40-binding epitope of CD40L. *In vitro* activated T cells from patients with CD40L deficiency fail to bind the soluble CD40-Fc (Fig. 2.5).

CD40 is expressed not only on B cells but also on the surfaces of macrophages, dendritic cells, follicular dendritic cells (FDCs), mast cells, and some epithelial

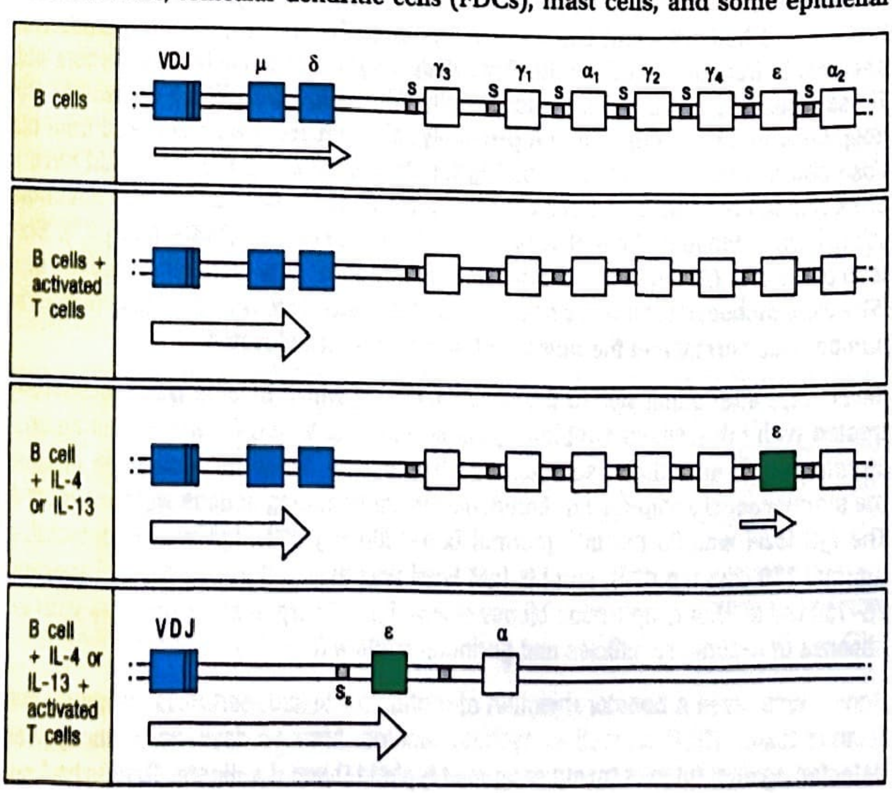

Fig. 2.4 Class switching to IgE production by human B cells. Purified human B cells in culture transcribe the μ and δ loci at a low rate, giving rise to surface IgM and IgD. On co-culture with T cells activated with ionomycin and phorbol myristate acetate (PMA), IgM is secreted. The presence of IL-4 or IL-13 stimulates an isotype switch to IgE. Purified B cells cultured alone with these cytokines transcribe the Cε gene at a low rate, but the transcripts originate in the switch region preceding the gene and do not code for protein. On co-culture with activated T cells in addition to IL-4 or IL-13, an isotype switch occurs, mature ε RNA is expressed, and IgE is synthesized.

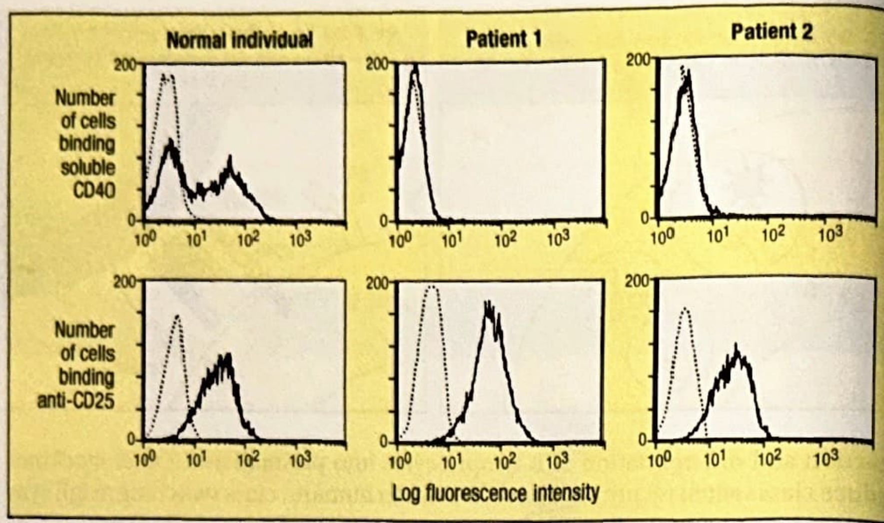

Fig. 2.5 Flow cytometric analysis showing that activated T cells from hyper IgM patients do not express the CD40 ligand. T cells from two patients and one healthy donor were activated *in vitro* with a T-cell mitogen, incubated with soluble CD40 protein, and analyzed by flow cytometry (see Fig. 1.3). The results are shown in the top three panels. In the normal individual, there are two populations of cells: one that does not bind CD40 (the peak to the left, with low-intensity fluorescence) and one that does (the peak to the right, with high-intensity fluorescence). The dotted line is the negative control, showing nonspecific binding of a fluorescently labeled protein to the same cells. In the patients with CD40L deficiency (center and right-hand panels), CD40 fluorescence exactly coincides with the nonspecific control, showing that there is no specific binding to CD40 by these cells. The bottom panels show that the T cells have been activated by the mitogen, because the T cells of both the normal individual and the two patients have increased expression of the IL-2 receptor (CD25), as expected after T-cell activation. The negative control is fluorescent goat anti-mouse immunoglobulin.

and endothelial cells. Macrophages and dendritic cells are antigen-presenting cell that can trigger the initial activation and expansion of antigen-specific T cells at th start of an immune response. Experiments in CD40L- or CD40-deficient patient and in gene-targeted CD40L-deficient mice indicate a role for the CD40–CD40 interaction in this early priming event, because in the absence of either CD40L o CD40 the initial activation and expansion of T cells in response to protein antigen is greatly reduced. The impairment of T-cell activation is the basis of some sever clinical features that distinguish CD40L and CD40 deficiency from other condi tions characterized by a pure antibody deficiency.

The case of Dennis Fawcett: a failure of T-cell help.

Five-year-old boy fails to make antibody against strep infection.

Dennis Fawcett was 5 years old when he was referred to the Children's Hospital wit a severe acute infection of the ethmoid sinuses (ethmoiditis). His mother reporte that he had had recurrent sinus infections since he was 1 year old. Dennis ha pneumonia from an infection with *Pneumocystis jirovecii* when he was 3 years old These infections were treated successfully with antibiotics. While he was in the hospital with ethmoiditis, group A β-hemolytic streptococci were cultured from his nose and throat. The physicians caring for Dennis expected that he would have a brisk rise in his white blood cell count as a result of his severe bacterial infection yet his white blood cell count was 4200 μl^{-1} (normal count 5000–9000 μl^{-1}). Sixteen percent of his white blood cells were neutrophils, 56% were lymphocytes, and 28% were monocytes. Thus his neutrophil number was low, whereas his lymphocyte number was normal and the number of monocytes was elevated.

Seven days after admission to the hospital, during which time he was successfully treated with intravenous antibiotics, his serum was tested for antibodies against streptolysin O, an antigen secreted by streptococci. When no antibodies against the streptococcal antigen were found, his serum immunoglobulins were measured. The IgG level was 25 mg dl^{-1} (normal 600–1500 mg dl^{-1}), IgA was undetectable (normal 150–225 mg dl^{-1}), and his IgM level was elevated at 210 mg dl^{-1} (normal 75–150 mg dl^{-1}). A lymph-node biopsy showed poorly organized structures with an absence of secondary follicles and germinal centers (Fig. 2.6).

Dennis was given a booster injection of diphtheria toxoid, pertussis antigens, and tetanus toxoid (DPT) as well as typhoid vaccine. After 14 days, no antibody was detected against tetanus toxoid or against typhoid O and H antigens. Dennis had red

blood cells of group O. People with type O red blood cells make antibodies against the A substance of type A red cells and antibodies against the B substance of type B red cells. This is because bacteria in the intestine have antigens that are closely related to A and B antigens. Dennis's anti-A titer was 1:3200 and his anti-B titer 1:800, both very elevated. His anti-A and anti-B antibodies were of the IgM class only.

His peripheral blood lymphocytes were examined by fluorescence-activated cell sorting analysis, and normal results were obtained: 11% reacted with an antibody against CD19 (a B-cell marker), 87% with anti-CD3 (a T-cell marker), and 2% with anti-CD56 (a marker for natural killer (NK) cells). However, all of his B cells ($CD19^+$) had surface IgM and IgD and none were found with surface IgG or IgA. When his T cells were activated *in vitro* with phorbol ester and ionomycin (a combination of potent polyclonal T-cell activators), they did not bind soluble CD40.

Dennis had an older brother and sister. They were both well. There was no family history of unusual susceptibility to infection.

Dennis was treated with intravenous gamma globulin, 600 mg kg^{-1} body weight each month, and subsequently remained free of infection until 15 years of age, when he developed severe, watery diarrhea. *Cryptosporidium parvum* was isolated from the stools. Within a few months, during which his diarrhea persisted in spite of treatment with the antibiotic azithromycin, he developed jaundice. His serum total bilirubin level was 8 mg dl^{-1}, and the serum level of conjugated bilirubin was 7 mg dl^{-1}. In addition, levels of γ-glutamyl transferase (γ-GT) and of the liver enzymes alanine aminotransferase (ALT) and aspartate aminotransferase (AST) were elevated at 93 IU l^{-1}, 120 IU μl^{-1}, and 95 IU μl^{-1}, respectively, suggesting cholestasis. A liver biopsy showed abnormalities of biliary ducts (vanishing bile ducts) that progressed to sclerosing cholangitis (chronic inflammation and fibrosis of the bile ducts). In spite of supportive treatment, Dennis died of liver failure at 21 years of age.

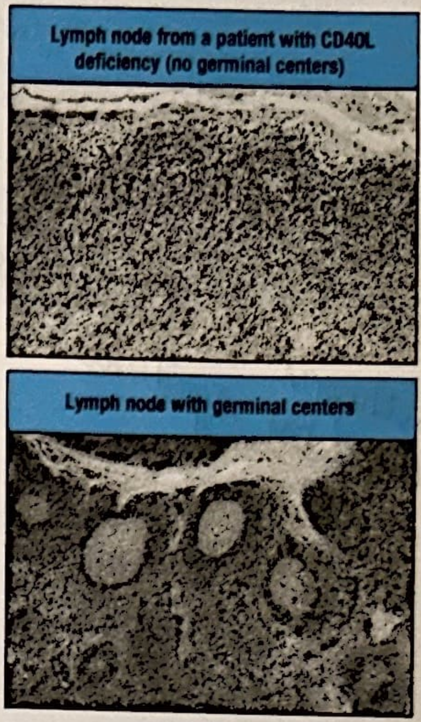

Fig. 2.6 Comparison of lymph nodes from a patient with CD40L deficiency (upper panel) and a normal individual (lower panel). Lower photograph courtesy of A. Perez-Atayde.

CD40 ligand deficiency (CD40L deficiency).

Males with a hereditary deficiency of CD40L exhibit consequences of a defect in both humoral and cell-mediated immunity. As we saw in Case 1, defects in antibody synthesis result in susceptibility to so-called pyogenic infections. These infections are caused by pyogenic (pus-forming) bacteria such as *Haemophilus influenzae*, *Streptococcus pneumoniae*, *Streptococcus pyogenes*, and *Staphylococcus aureus*, which are resistant to destruction by phagocytic cells unless they are coated (opsonized) with antibody and complement. On the other hand, defects in cellular immunity result in susceptibility to opportunistic infections. Bacteria, viruses, fungi, and protozoa that often reside in our bodies and only cause disease when cell-mediated immunity in the host is defective are said to cause opportunistic infections.

Dennis revealed susceptibility to both kinds of infection. His recurrent sinusitis, as we have seen, was caused by *Streptococcus pyogenes*, a pyogenic infection. He also had pneumonia caused by *Pneumocystis jirovecii* and diarrhea caused by *Cryptosporidium parvum*, a fungus and a protozoan, respectively, that are ubiquitous and cause opportunistic infections in individuals with defects in cell-mediated immunity.

Patients with a CD40L deficiency can make IgM in response to T-cell independent antigens, but they are unable to make antibodies of any other isotype, and they cannot make antibodies against T-cell dependent antigens, which leaves the patient largely unprotected from many bacteria. They also have a defect in cell-mediated immunity that strongly suggests a role for CD40L in the T cell-mediated activation of macrophages. *Cryptosporidium* infection can cause persistent inflammation in the liver, and ultimately sclerosing cholangitis and liver failure. In addition, individuals with CD40L deficiency have severe neutropenia, with a block at the promyelocyte/myelocyte stage of differentiation in the bone marrow. Although the mechanisms underlying the neutropenia in these patients remain unclear, the lack of neutrophils accounts for the presence of severe sores and blisters in the mouth. The neutropenia and its consequences can often be overcome by administering recombinant granulocyte-colony stimulating factor (G-CSF).

Septrin
• Trimethoprim + sulfamethoxazole
are the active ingredients.

Treatment of CD40L deficiency is based on immunoglobulin replacement therapy, prophylaxis with trimethoprim-sulfamethoxazole to prevent *Pneumocystis jirovecii* infection, and protective measures to reduce the risk of *Cryptosporidium* infection (such as avoiding swimming in lakes or drinking water with a high concentration of *Cryptosporidium* cysts). In spite of this, many patients with CD40L deficiency die in late childhood or adulthood of infections, liver disease, or tumors (lymphomas and neuroectodermal tumors of the gut). The disease can be cured by hematopoietic cell transplantation, and this treatment should be considered when HLA-identical donors are available and when the first signs of severe complications become manifest.

Few cases of CD40 deficiency have been reported. Its clinical and immunological features, and its treatment, are very similar to CD40L deficiency, but the disease is inherited as an autosomal recessive trait.

Questions.

1 *Dennis's B cells expressed IgD as well as IgM on their surface. Why did he not have any difficulty in isotype switching from IgM to IgD?*

2 *Normal mice are resistant to Pneumocystis jirovecii. SCID mice, which have no T or B cells but have normal macrophages and monocytes, are susceptible to this microorganism. In normal mice, Pneumocystis jirovecii organisms are taken up and destroyed by macrophages. Macrophages express CD40. When SCID mice are reconstituted with normal T cells they acquire resistance to Pneumocystis infection. This can be abrogated by antibodies against the CD40 ligand. What do these experiments tell us about this infection in Dennis?*

3 *Why did Dennis make antibodies against blood group A and B antigens but not against tetanus toxoid, typhoid O and H, and streptolysin antigens? Would he have made any antibodies in response to his Streptococcus pyogenes infection?*

4 *Most IgM is circulating in the blood, and less than 30% of IgM molecules get into the extravascular fluid. On the other hand, more than 50% of IgG molecules are in the extravascular space. Furthermore, we have 30–50 times more IgG than IgM in our body. Why are IgG antibodies more important in protection against pyogenic bacteria?*

5 *Newborns have difficulty in transcribing the gene for CD40L. Does this help to explain the susceptibility of newborns to pyogenic infections? Cyclosporin A, a drug widely used for immunosuppression in graft recipients, also inhibits transcription of the gene for CD40L. What does this imply for patients taking this drug?*